Week 7

1. Random Forest is best described as:

A) A single deep decision tree
B) A clustering algorithm
C) A neural network ensemble
D) A bagging of de-correlated decision trees

Answer: D

Explanation

A. A single deep decision tree (Incorrect): Random Forest consists of many decision trees, not just one deep tree.

B. A clustering algorithm (Incorrect): Random Forest is primarily used for classification and regression, whereas clustering is an unsupervised learning task.

C. A neural network ensemble (Incorrect): Random Forest is based on decision trees and bagging, not neural networks.

D. A bagging of de-correlated decision trees (Correct): Random Forest builds multiple decision trees using bootstrap samples and random feature selection, then combines their predictions to improve accuracy and reduce overfitting.

2. What is the main purpose of Bagging?

A) Increase bias
B) Reduce prediction variance
C) Reduce dataset size
D) Increase overfitting

Answer: B

Explanation

A. Increase bias (Incorrect): Bagging is mainly used to reduce variance, not to increase bias.

B. Reduce prediction variance (Correct): Bagging (Bootstrap Aggregating) trains multiple models on different bootstrap samples and combines their predictions, which helps reduce variance and improve model stability.

C. Reduce dataset size (Incorrect): Bagging creates multiple sampled datasets but does not reduce the original dataset size.

D. Increase overfitting (Incorrect): Bagging generally helps reduce overfitting by averaging predictions from multiple models.

3. Bootstrap in Bagging refers to:

A) Sampling with replacement
B) Feature scaling
C) Gradient computation
D) Data normalization

Answer: A

Explanation

A. Sampling with replacement (Correct): Bootstrap sampling creates multiple training datasets by randomly selecting instances from the original dataset with replacement, meaning the same instance can appear multiple times in a sample.

B. Feature scaling (Incorrect): Feature scaling adjusts the range of features and is unrelated to bootstrap sampling.

C. Gradient computation (Incorrect): Gradient computation is used in optimization algorithms such as gradient descent, not in bagging.

D. Data normalization (Incorrect): Data normalization standardizes feature values but is not the process referred to as bootstrap in bagging.

4. In Random Forest, at each split:

A) All features are always evaluated
B) Random subset of features is selected
C) Features are ignored
D) Only one feature exists

Answer: B

Explanation

A. All features are always evaluated (Incorrect): Evaluating all features at every split is done in standard decision trees, not Random Forest.

B. Random subset of features is selected (Correct): At each node split, Random Forest considers only a random subset of available features, which helps create diverse trees and reduces correlation among them.

C. Features are ignored (Incorrect): Features are actively used for splitting; Random Forest simply selects a random subset of them.

D. Only one feature exists (Incorrect): Random Forest typically works with datasets containing multiple features and chooses from a random subset at each split.

5. **Random Forest can naturally support parallel processing because:**

A) Trees are built independently
B) Trees share memory
C) Trees use one processor
D) Trees require sequential execution

Explanation

Answer: A

A. Trees are built independently (Correct): Each decision tree in a Random Forest is trained separately on its own bootstrap sample, allowing multiple trees to be built simultaneously on different processors or machines.

B. Trees share memory (Incorrect): Trees do not need to share memory during training; they operate independently.

C. Trees use one processor (Incorrect): Random Forest can utilize multiple processors to train trees in parallel, improving efficiency.

D. Trees require sequential execution (Incorrect): Unlike some algorithms that depend on previous steps, Random Forest trees can be trained concurrently because they are independent of each other.

6. Boosting combines:

A) Strong classifiers only
B) Weak classifiers into a strong ensemble
C) Clusters into datasets
D) Databases into warehouses

Answer: B

Explanation

A. Strong classifiers only (Incorrect): Boosting is specifically designed to improve the performance of weak learners rather than combining already strong classifiers.

B. Weak classifiers into a strong ensemble (Correct): Boosting sequentially combines multiple weak learners, where each new model focuses on correcting the errors of previous ones, resulting in a strong predictive model.

C. Clusters into datasets (Incorrect): Clustering groups similar data points and is unrelated to the boosting process.

D. Databases into warehouses (Incorrect): Data warehousing is a database management concept and has no connection with boosting in machine learning.

7. Gradient Boosting builds models:

A) In a single step
B) Iteratively
C) Randomly
D) Without training

Answer: B

Explanation

A. In a single step (Incorrect): Gradient Boosting requires multiple iterations to gradually improve performance.

B. Iteratively (Correct): Gradient Boosting builds models sequentially, where each new model is trained to correct the errors (residuals) of the previous models.

C. Randomly (Incorrect): Although some randomness may be added in variants, the core Gradient Boosting process is a systematic iterative procedure.

D. Without training (Incorrect): Each weak learner must be trained on the data, so training is an essential part of the algorithm.

8.In Gradient Boosted Trees, the base learners are usually:

A) Decision Trees
B) Neural Networks
C) K-Means Clusters
D) Support Vectors

Answer: A

Explanation

A. Decision Trees (Correct): Gradient Boosted Trees use a sequence of small decision trees (weak learners), where each tree attempts to correct the errors made by the previous trees.

B. Neural Networks (Incorrect): Neural networks are a different type of machine learning model and are not the standard base learners in Gradient Boosted Trees.

C. K-Means Clusters (Incorrect): K-Means is a clustering algorithm and cannot serve as the weak learner in Gradient Boosting.

D. Support Vectors (Incorrect): Support Vector Machines are classification/regression models, but they are not the typical base learners used in Gradient Boosted Trees.

9. Cross-validation is mainly used to:

A) Increase dataset size
B) Assess model quality and select best parameters
C) Reduce number of features
D) Perform clustering

Answer: B

Explanation

A. Increase dataset size (Incorrect): Cross-validation reuses existing data for training and testing; it does not create new data.

B. Assess model quality and select best parameters (Correct): Cross-validation evaluates how well a model generalizes to unseen data and helps in selecting the optimal model and hyperparameters.

C. Reduce number of features (Incorrect): Feature reduction is performed using feature selection or dimensionality reduction techniques, not cross-validation.

D. Perform clustering (Incorrect): Clustering is an unsupervised learning task, whereas cross-validation is primarily used for model evaluation and selection.

10. Recommended learning rate for Gradient Boosting is:

A) Greater than 1
B) Equal to 5
C) Less than or equal to 0.1
D) Equal to 10

Answer: C

Explanation

A. Greater than 1 (Incorrect): Learning rates greater than 1 usually cause unstable training and poor model performance.

B. Equal to 5 (Incorrect): A learning rate of 5 is excessively high and would likely prevent the model from converging properly.

C. Less than or equal to 0.1 (Correct): A small learning rate (commonly 0.1 or lower) helps Gradient Boosting learn gradually, improving generalization and reducing the risk of overfitting.

D. Equal to 10 (Incorrect): Such a large learning rate makes updates too aggressive, often leading to divergence rather than improved learning.